

Course Handout

Institute/School	Chitkara University Institute of Engineering and Technology		
Department	Department of Applied Sciences		
Programme	Bachelor of Engineering (B.E.), Computer Science & Engineering		
Course Title	Linear Algebra and Calculus	Session	July-Dec 2026
Course Code	26APS1101	Semester/Batch	1 st /2026
L-T-P (Per Week)/No. of Hours (Per Week)	4-0-0	Course Credits	04
Pre-requisite	NA	NHEQF Level	4.5
Course Coordinator	Dr. Swati	SDG Number	4, 9

1. Scope and objectives of the Course

This course deals with the ability to apply mathematics to provide the innovative solutions of complex engineering and real-life problems in a qualitative manner. It helps to identify, formulate and analyze the engineering problems which come and to arrive at substantiated conclusions by using principles of mathematics. The main objectives of the course are:

- To analyze functions of two or more variables and compute their derivatives for finding extreme values of surfaces.
- To design matrices for mathematical problems related to real life and find their solutions.
- To apply techniques of evaluating double and triple integral to solve various engineering problems.
- To compute gradient, divergence, and curl and evaluate line, surface and volume integrals using Green's, Stokes' and Gauss divergence theorems for solving engineering problems.

2. Course Outcomes (COs)

After completing the course, students should be able to:

CO1: Find local extreme values of functions of several variables, test for saddle points, examine the conditions for the existence of absolute extreme values. Solve constraint problems using Lagrange multipliers and solve related application problems.

CO2: Form matrices to present mathematical solutions in a concise and informative manner. Use matrices to solve the problems of linear equations and solve various live problems using matrices.

CO3: Apply the principles of integral calculus to solve a variety of practical problems in engineering and applied sciences.

CO4: Apply multivariable vector-valued functions, their derivatives, and integrals to live problems, graphically and analytically.

CO-PO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	No. of Lectures	NHEQF Level Descriptor
CO1	H	H	M	L								H	M	13	Q1, Q3, Q4, Q5, Q6
CO2	H	H	M	M	M							H	M	11	Q1, Q2, Q3, Q4, Q6
CO3	H	H	M	L	M							M	L	17	Q1, Q3, Q4, Q5, Q6
CO4	H	H	M	M	M							M	M	19	Q1, Q4, Q5, Q6
Total Contact Hours														60	

H=High, M=Medium, L=Low

3. Recommended Books/E-Resources (Textbooks and Reference books):

- B01:** Higher Engineering Mathematics, B.S. Grewal, Khanna Publications, 45th Edition.
B02: Advanced Engineering Mathematics, Erwin Kreyszig, Wiley India Pvt. Ltd, 10th Edition.
B03: Engineering Mathematics, Srimanta Pal & Subodh C. Bhunia, Oxford University Press, 1st Edition.
B04: The Engineering Mathematics, Sita Ram et.al., Chitkara University Publication, Vol. I, 2nd Edition.
B05: Higher Engineering Mathematics, B.V. Ramana, Tata McGraw-Hill Education, 6th Edition.
B06: Advanced Engineering Mathematics, R.K. Jain and S.R.K. Iyengar, Alpha Science International Ltd 5th Edition.
B07: A Textbook of Engineering Mathematics, N. P. Bali and Manish Goyal, Laxmi Publications, 9th Edition
B08: Calculus, Howard Anton, Irl Bivens Stephens Davis, John Wiley & Sons Inc, 10th Edition
B09: Advanced Engineering Mathematics, H.C. Taneja, I.K. International Publishing House Pvt. Ltd, 2nd Edition.

E-Library Resource

<https://library.chitkara.edu.in/subscribed-books.php>

4. Recommended Tools and Platforms

C/ C++, Python, MATLAB.

5. Delivery/Instructional Plan and Resources

6. Theory Plan:

Lect. No.	Topics	CO	Book No, CH. No	TLM	Web References/Audio-Video
1-4	Partial Derivatives: Partial Differentiation, Euler's Theorem for Homogeneous Functions.	CO1	B01, CH 5	Lecture	https://nptel.ac.in/courses/1111051 21
5-7	Tangent and Normal, Partial Derivative of Composite Functions, Implicit Functions.	CO1	B01, CH 5	Lecture	https://nptel.ac.in/courses/1111051 21
8-9	Total Derivatives, Errors & Approximation.	CO1	B01, CH 5	Lecture	https://nptel.ac.in/courses/1111051 21
10-11	Jacobians with Properties, Taylor's Series Expansion, Maclaurin's Series (One & Two Variables).	CO1	B01, CH 5	Lecture	https://nptel.ac.in/courses/1221040 17
12-13	Maxima and Minima of Functions of Two and Three Variables, Lagrange's Method of Undetermined Multipliers.	CO1	B01, CH 5	Lecture	https://nptel.ac.in/courses/1221040 17
14-17	Matrices: Rank, Inverse of Matrix, Normal Form.	CO2	B01, CH 1	Lecture	https://nptel.ac.in/courses/1111011 15
18-21	Solution of a System of Linear Equations by Using Rank, Characteristics Equations, Eigen Values and Eigen Vectors.	CO2	B01, CH 1	Lecture	https://nptel.ac.in/courses/1111011 15
22-24	Cayley Hamilton theorem (without proof), Diagonalization, Quadratic form & Canonical form.	CO2	B01, CH 1	Lecture	https://nptel.ac.in/courses/1111011 15
25-27	Curve tracing (Cartesian and Polar Curves): Cissoid, Cardioid, Lemniscate, Folium of Descartes, Three/Four Leaved Rose, Limaçon.	CO3	B01, CH 6	Lecture	https://nptel.ac.in/courses/1221040 17
28-33	Multiple Integral: Double Integration using Cartesian & Polar Coordinates, Change of Order in Double Integration,	CO3	B01, CH 7	Lecture	https://nptel.ac.in/courses/1111051 21

	Change of Variables in Polar.				
34-37	Cylindrical and Spherical Coordinates, Area Enclosed by Plane curves, Triple Integration, Volume.	CO3	B01, CH 7	Lecture	https://nptel.ac.in/courses/111105121
38-41	Moment of Inertia, Centroid and Center of Gravity, Improper Integrals of First and Second kind, Beta and Gamma functions.	CO3	B01, CH 7	Lecture	https://nptel.ac.in/courses/111105121
42-44	Vector Calculus: Scalar and Vector Field, Gradient, Directional Derivatives. Divergence, Curl, Properties of Curl, gradient, Divergence	CO4	B02, CH 8	Lecture	https://nptel.ac.in/courses/122101002
45-47	Line Integral, Surface and Volume Integral.	CO4	B02, CH 8	Lecture	https://nptel.ac.in/courses/122104017
48-60	Green's Theorem in the Plane (without proof) and applications, Stoke's Theorem (without proof) and applications, Gauss Divergence Theorem (without proof) and applications.	CO4	B02, CH 8	Lecture	https://nptel.ac.in/courses/122104017

7. **Course ERISE Mapping:** Entrepreneurship/Research/Innovation/Ability Enhancement/ ☒ Skill Enhancement/Employability

8. **Active Learning Methodology Used**

- **Matrices (Rank, Eigenvalues, Diagonalization): Think–Pair–Share and Group Discussions** were used to solve numerical problems, compare solution approaches, and relate concepts to engineering applications.
- **Partial Derivatives & Optimization: Quiz/Test Questions and Think–Pair–Share** were used to strengthen conceptual understanding and problem-solving skills.
- **Curve Tracing, Multiple Integrals & Vector Calculus: Group Discussions and Quiz/Test Questions** promoted analytical thinking and application of concepts to real-world engineering problems.

9. **Action plan for different types of learners**

Fast Learners	Average Learners	Learners with constant monitoring
• Advanced practice questions and challenge problems • Mini projects or presentations on course topics • Peer tutoring to help classmates • Additional reference materials and online resources	• E-notes and practice worksheets for self-study • Participation in group discussions • Formative Exercises used to highlight concepts and notions • Audio-visual learning resources for better understanding	• Remedial Classes • Encouragement for improvement using Peer Tutoring • Regular monitoring with short practice tasks and feedback • Audio-visual learning resources for better understanding

10. **Additional Self-directed Learning**

Assignments to promote self-learning, survey of contents from multiple sources.

S. No	Advanced Topics, Additional Reading, Research papers and any	CO	ALM	References/MOOCs
1.	Implementation of Linear Algebra using C/Python.	CO2	Think – Pair- Share, Group Discussions	https://numpy.org/doc/stable/reference/routines.linalg.html
2.	Implementation of multiple Integral and vector calculus using C/Python.	CO3, CO4	Think – Pair- Share, Group Discussions	https://www.youtube.com/watch?v=-SdIZHPuW9o

11. Delivery Details of Content Beyond Syllabus:

- ☐ C/Python implementation of Linear Algebra and Vector Calculus.
- ☐ Visualization of vector fields and multiple integrals using C/Python.
- ☐ Industry expert session on the applications of Linear Algebra and Calculus in Computer Science.

12. Evaluation Scheme & Components:

Assessment Type	No. of Assessments	Type of Component	Component Description	% Weightage of Component	Max. Marks	Mode of Assessment	CO
Formative	05	Continuous Assessments	CAT1	25%	25	Online	CO1- CO3
			CAT2	25%	25	Online	
			CAT3	25%	25	Online	
			CAT4/CAT5*	25%	25	Online	CO1- CO4
	Total		100%				

*Appearing in either CAT4 or CAT5 is compulsory. Non-appearance in both CAT4 and CAT5 will automatically disqualify the student from clearing the subject.

The final score will be calculated based on the **best four out of all the five assessments.

***Students must secure a minimum aggregate of **50% marks** to pass a subject. Scoring below 50% will result in a **Fail grade (F)**.

13. Sessional Test-Wise Syllabus of the Course:

Evaluation Component	Description	Syllabus Covered	Weightage %
Continuous Assessment	CAT1	Lecture (1-11)	25%
	CAT2	Lecture (12-24)	25%
	CAT3	Lecture (25-37)	25%
	CAT4	Lecture (1-60)	25%
	CAT5	Lecture (1-60)	25%

* Minimum 75% attendance is required to become eligible for appearing in CAT 4 and CAT 5.

This Document is approved by:

Designation	Name	Signature with date
Prepared by Course Coordinator	Dr. Swati	
Verified by Program In charge (Department of Applied Sciences)	Prof. (Dr) Reetu Malhotra	
Approved by Dean (Department of Applied Sciences)	Prof. (Dr) Mohit Kumar Kakkar	
Date (DD/MM/YYYY)	14/08/2026	